

Service Level Agreements (Scenario A)

Slice	SD	Applications (gentle profiles)	DL Target (bps)	PRB Ceiling
CRITICAL	3	Robot 1 (Jetson) generic UDP traffic, no scripted apps	50,000	80%
PERFORMANCE	1	Robot 2 (Jetson) generic UDP traffic, no scripted apps	300,000	50%
BUSINESS	2	Google Pixel organic browsing / (unscripted)	0 (no target)	30%

Scenario Used in the Simulations (Scenario A)

Scenario	UEs per Slice	DUs / RIC	Bandwidth	Traffic
3 Physical UEs (2 Jetson robots + 1 Pixel)	CRITICAL: 1 (Robot 1, IMSI 005) PERFORMANCE: 1 (Robot 2, IMSI 004) BUSINESS: 1 (Pixel, IMSI 003)	1 RF DU (USRP X310), 1 RIC, 1 CU	5 MHz / 10 MHz (two runs)	Robots: generic UDP. Pixel: unscripted real-world browsing/streaming

Service Level Agreements (Scenario B)

Slice	SD	Applications	DL Target (bps)	PRB Ceiling
CRITICAL	3	Robot 1 (physical, IMSI 005) · vUE1 (virtual, IMSI 006) ato gentle profile (1/0.25 kbps, not video_surv)	50,000	80%
PERFORMANCE	1	Robot 2 (physical, IMSI 004) · vUE2 (virtual, IMSI 007)	300,000	50%
BUSINESS	2	Pixel (physical, IMSI 003) unscripted · vUE3 (virtual, IMSI 008) "random traffic" generator	0 (no target)	30%

Scenario Used in the Simulations (Scenario B)

Scenario	UEs per Slice	DUs / RIC	Bandwidth	Traffic
6 UEs (3 physical + 3 virtual)	CRITICAL: 2 (Robot 1, vUE1) PERFORMANCE: 2 (Robot 2, vUE2) BUSINESS: 2 (Pixel, vUE3)	RF DU (X310) + ZMQ DU, 1 shared RIC, 1 CU	10 MHz	Physical: as Scenario A. Virtual: gentle critical/performance profiles + unscripted "random" business

Service Level Agreements (Scenario C)

Slice	SD	Applications	DL Target (bps)	PRB Ceiling
CRITICAL	3	ETCS · Voice · Remote Engine Control · ato (video_surv stress, 300 kbps DL)	350,000	80%
PERFORMANCE	1	Telemetry NC · Equipment Control · Asset Telemetry · PIS · video_surv	300,000	50%
BUSINESS	2	Passenger WiFi (PWifi)	20,000,000	30%

Scenario Used in the Simulations (Scenario C)

Scenario	TOBA per Slice	DUs / RIC	Apps across TOBAs	Total Apps
12 Virtual UEs (1 UE = 1 TOBA gateway)	CRITICAL: 5 (vue1, 4, 7, 10, 12) PERFORMANCE: 4 (vue2, 5, 8, 11) BUSINESS: 3 (vue3, 6, 9)	4 ZMQ DUs (3 UEs each) 1 shared RIC, 1 CU	5× critical (voice, etcs, ato, remote_engine_ctrl, pub_warn) 4× performance (telemetry_nc, equip_ctl, asset_tel, pis, video_surv) 3× business (passenger_wifi)	12